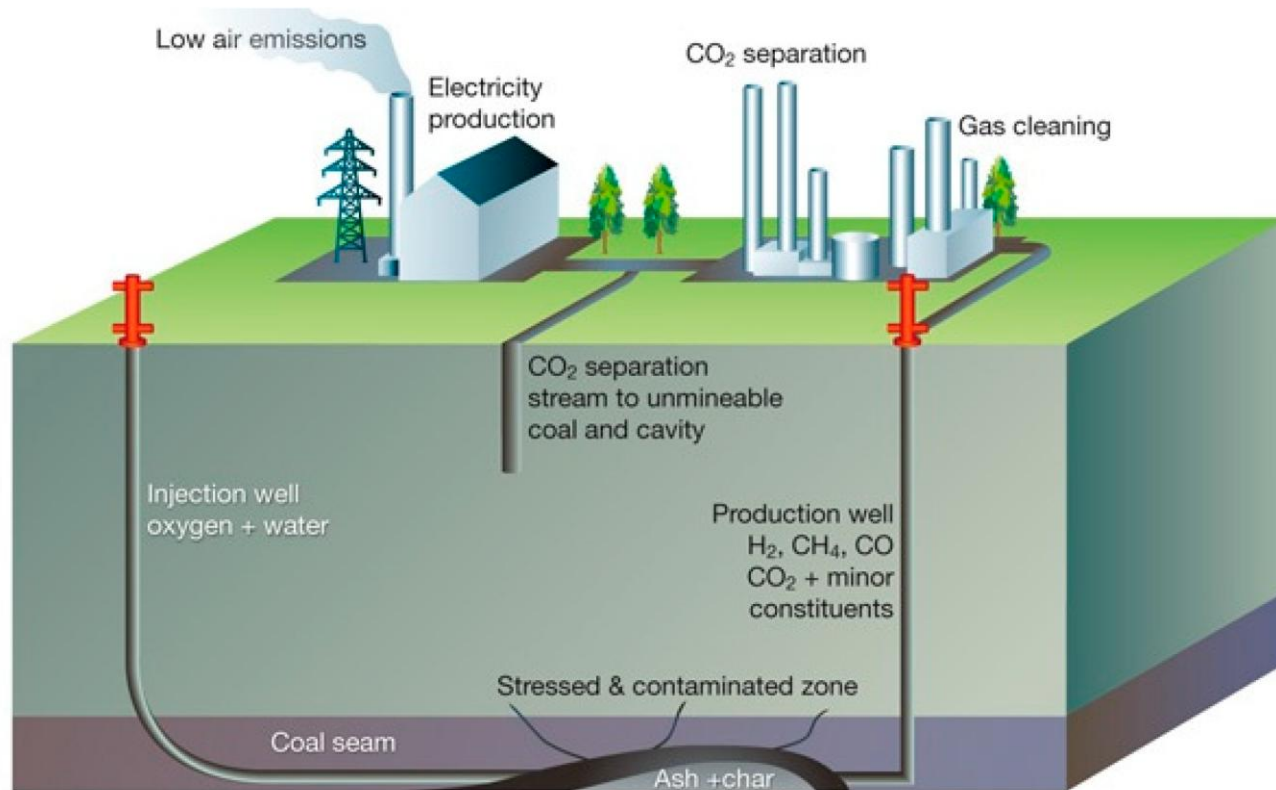


Underground Coal Gasification Method



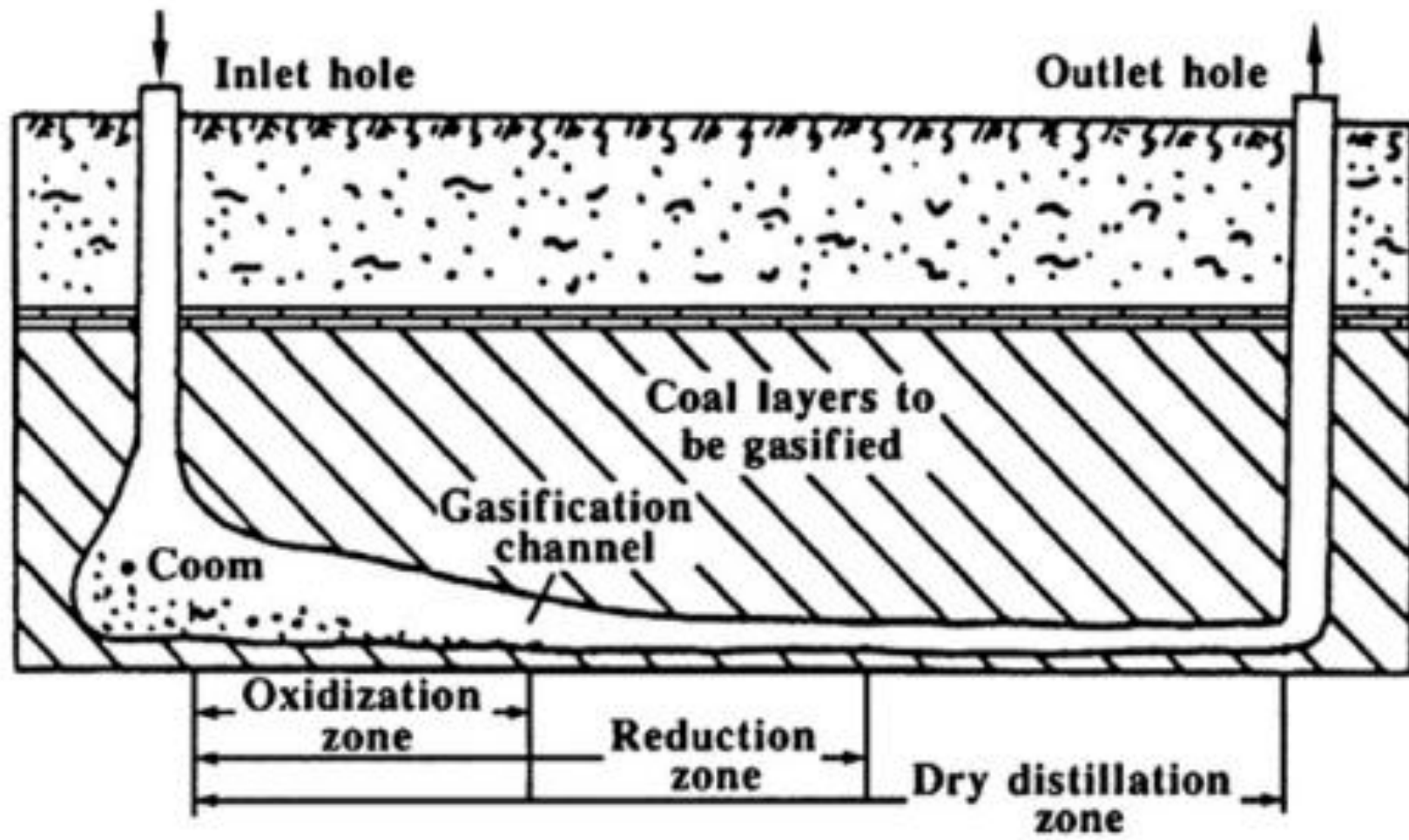
Why UCG ?

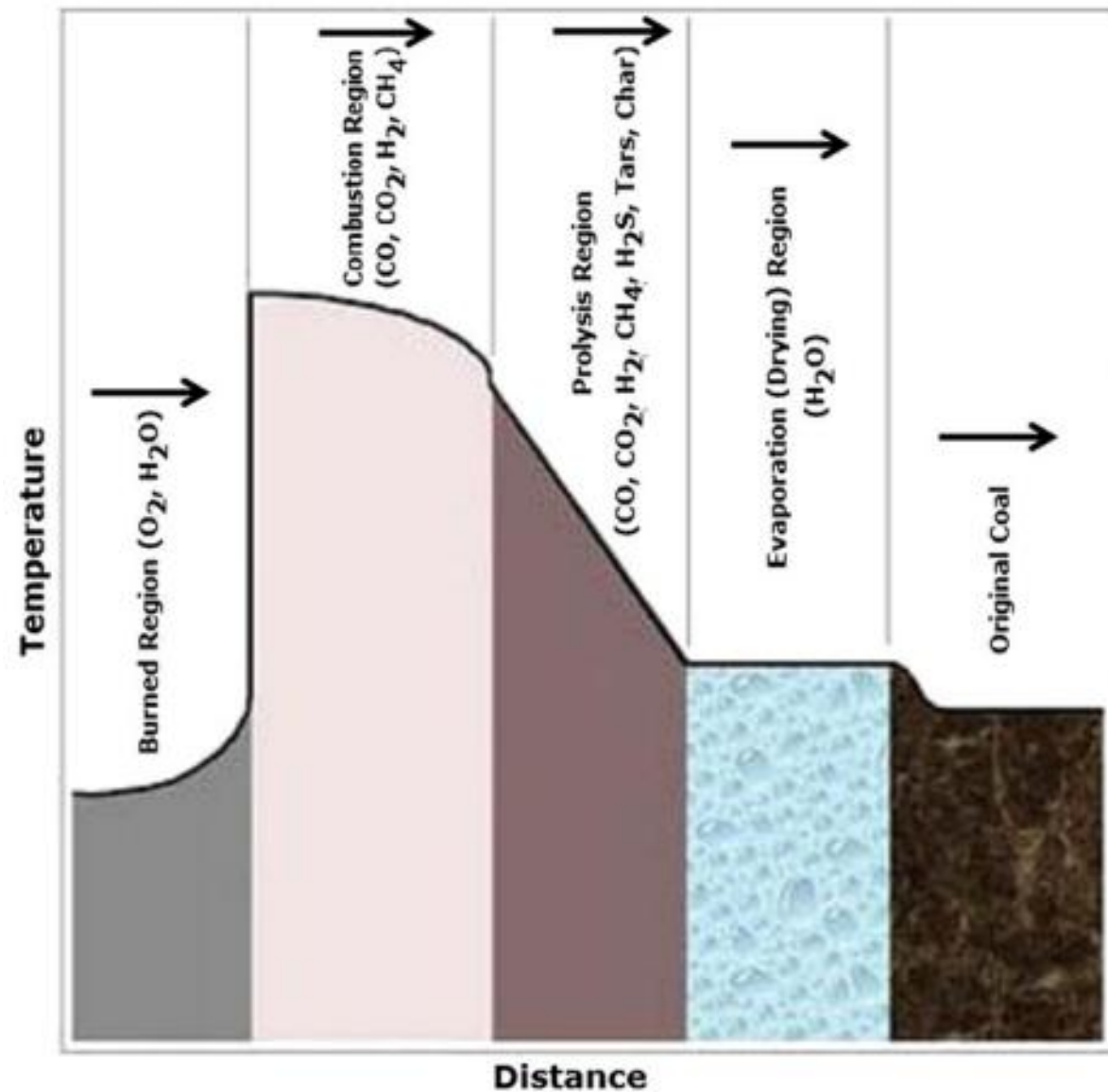
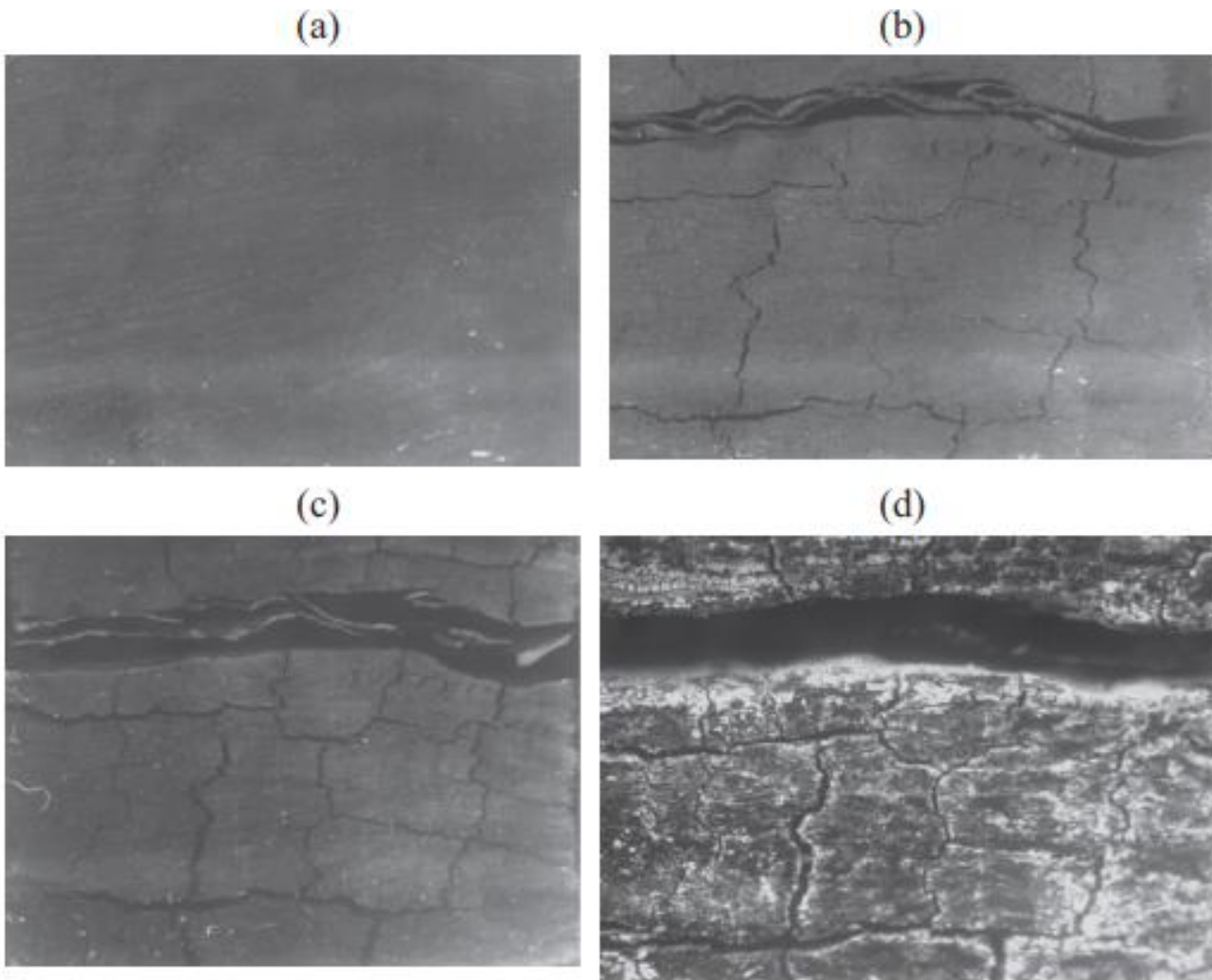
- Significant volume of access to deep lignite/sub-bituminous coals in India
- Reduced environmental impacts
- Enhanced efficiency and energy yield

- Underground coal gasification(UCG) enables coal to be converted into product/syn-gas.
- The coal seams that are **uneconomic, unviable, and inaccessible for traditional mining methods** can be utilized for underground coal gasification (UCG) process.
- Additionally, Non-coking coal poses some unique properties of **high reactivity, spontaneous combustibility, high moisture and low carbon contents**, which advocates to choose UCG method, wherever applicable.

Site Attributes for UCG

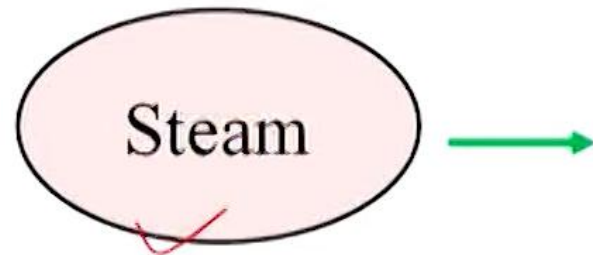
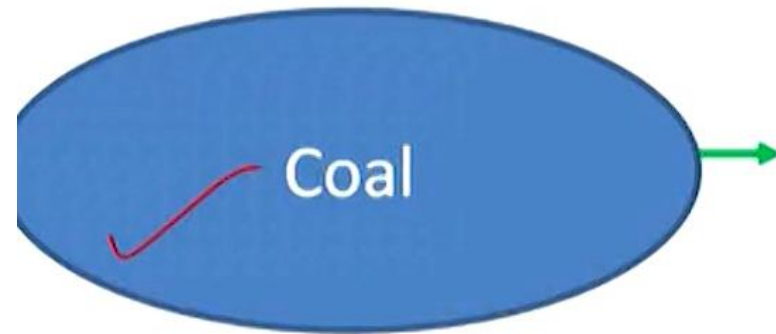
ATTRIBUTE	REQUIREMENT
COAL DEPTH	GREATER (> approx. 300 m)
COAL THICKNESS	ECONOMICALLY VIABLE (> approx. 6 m)
COAL RANK	LIGNITE TO SUB-BITUMINOUS
CAP ROCK	IMPERMEABLE STRATA SHOULD OVERLAY AND UNDERLAY (ex. CLAYSTONE)
PRESSURE	LOWER PRESSURE INSIDE THE GASIFICATION CHAMBER THAN THE SURROUNDING
FAULT/FRACTURES	LIMITED NUMBER OF FAULTS & FRACTURES
COMPACTION & SUBSIDENCE	NEGLECTIBLE
WATER QUALITY	POOR (NOT DRINKABLE)



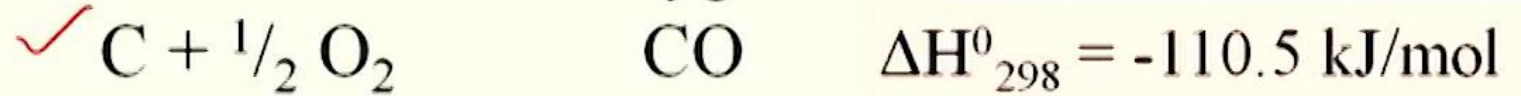


Thermal wave propagation through coal seam during in situ gasification

Basic chemistry of coal gasification



Gasification with Oxygen



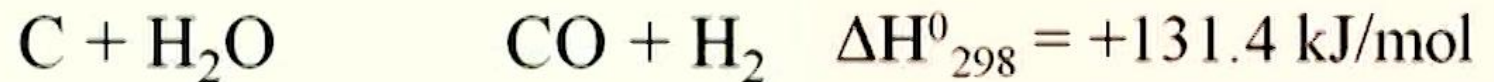
Combustion with Oxygen



Gasification with Carbon Dioxide (Boudouard reaction)



Gasification with Steam



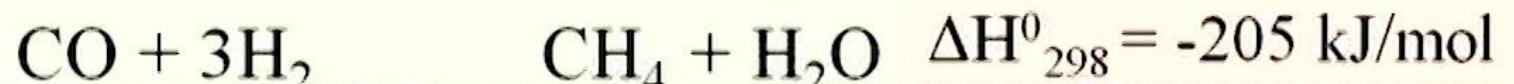
Gasification with Hydrogen



Water-Gas Shift

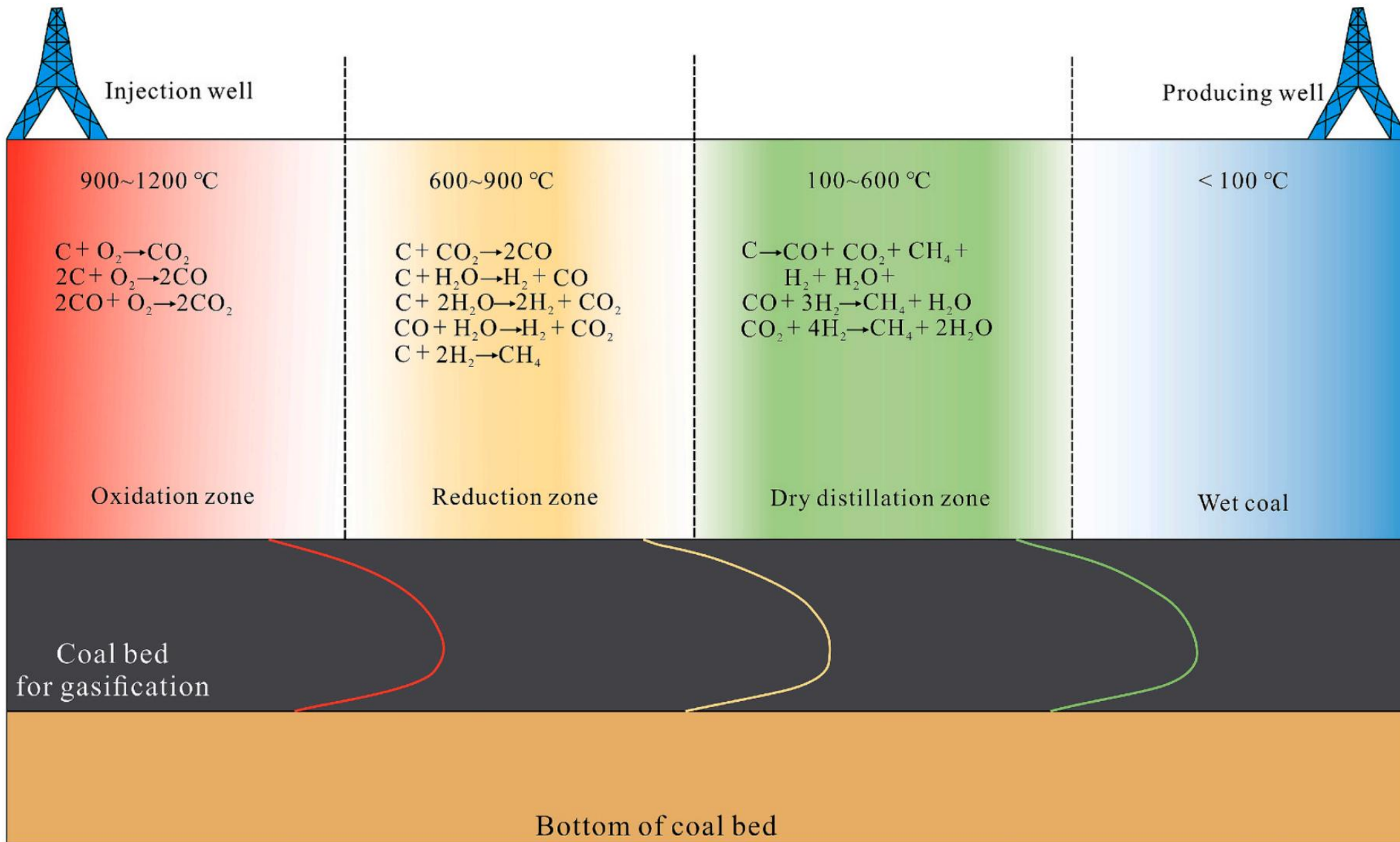


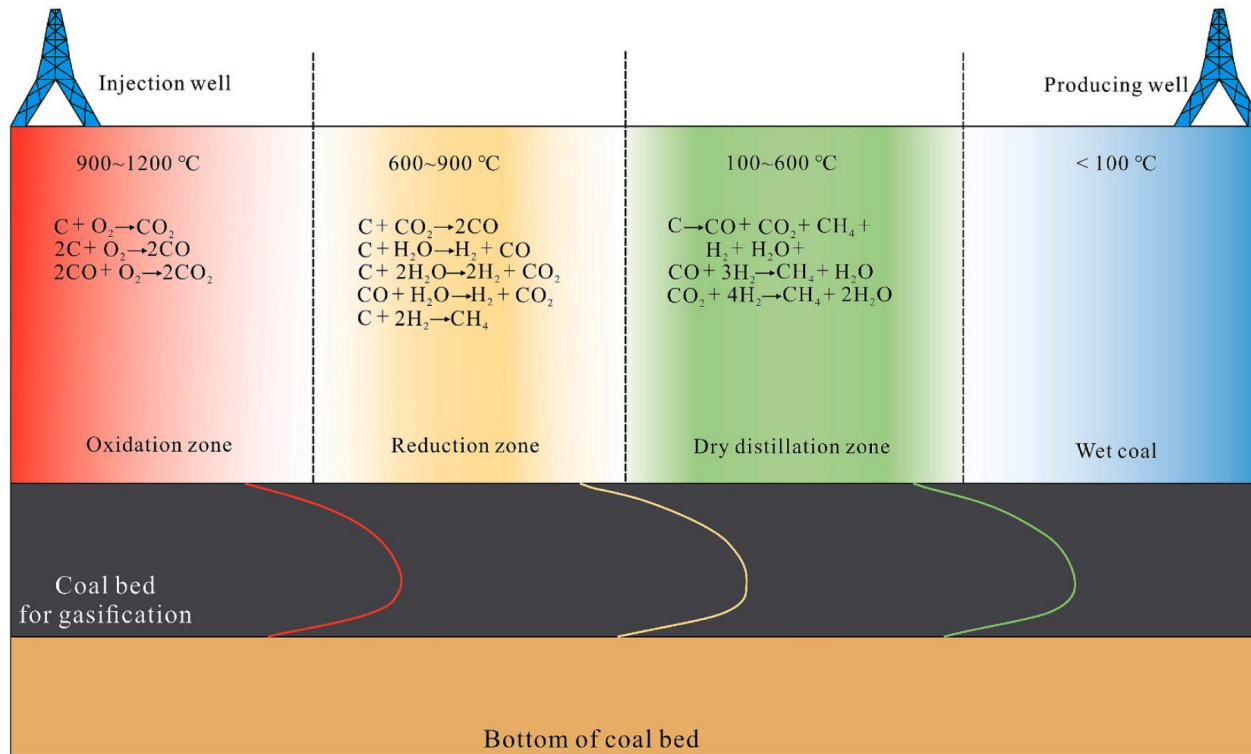
Methanation



Underground coal gasification

- UCG permits coal to be **gasified in situ** within the coal seam, via a matrix of wells. The coal is **ignited and air is injected** underground to sustain a fire, which is essentially used to **produce and transport combustible synthetic gas** to surface.
- This synthetic gas can be used for **industrial heating, power generation** or the **manufacture of hydrogen, synthetic natural gas or other fuels**.
- As compared with conventional mining and surface gasification, **UCG promises lower capital/operating costs** and also has other advantages, such as **no human labor underground** for coal mining.
- The **increasing demand for energy, depletion of oil and gas resources**, and **threat of global climate change** have led to growing interest in UCG throughout the world.
- The primary components of UCG syngas are **H₂, CO, CO₂, CH₄, and H₂S**.



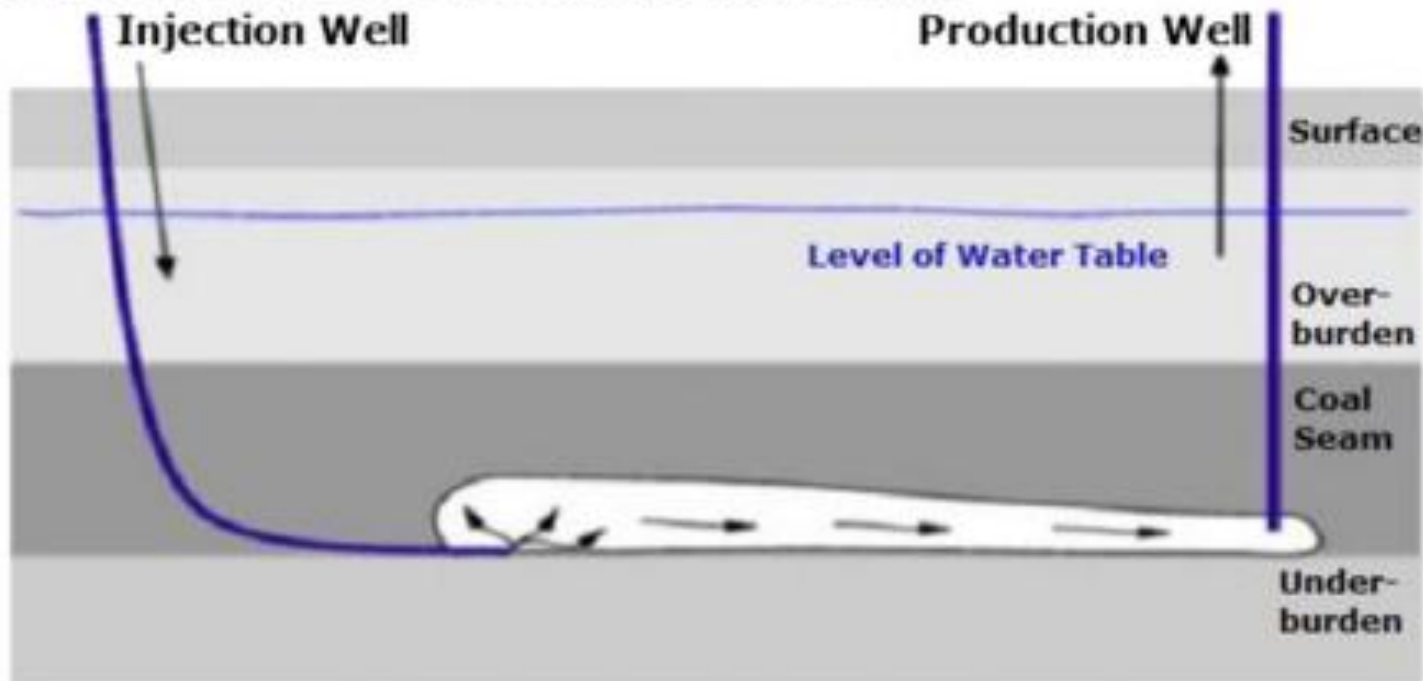


- (1) In the oxidation zone, O_2 from the gasification agent injected into the channel reacts with carbon (C) in the coal seam through complete and incomplete oxidation, producing CO_2 and CO while releasing substantial heat. This heat affects the coal seam, elevating the temperature of carbon in the oxidation zone and adjacent areas to as high as 1200 °C. During this process, O_2 concentration declines while CO_2 and CO concentrations rise, with oxidation ceasing when O_2 levels approach zero.
- (2) Reduction Zone: Following the airflow from the oxidation zone, CO_2 from the oxidation zone reacts with hot coal in the reduction zone to produce $CO \cdot H_2O(g)$ in the gasification agent reacts with hot coal, forming CO and H_2 through reduction. This endothermic process lowers the reduction zone's temperature. The reduction reaction ceases when the temperature drops below the required threshold, typically ranging from 600 to 1200 °C.
- (3) Dry Distillation Zone: After the reduction zone, the still-high airflow temperature affects the downstream coal seam, triggering volatilization and pyrolysis of gases, followed by methanation at 200–600 °C.

- After coal sequentially passes through the oxidation, reduction, and dry distillation and drying zones, its gasification products primarily consist of a combustible gas mixture comprising CO, H_2 , CH_4 , and other components.
- As the gasification front advances, the reaction progressively converts solid coal in the seam entirely into gaseous products.

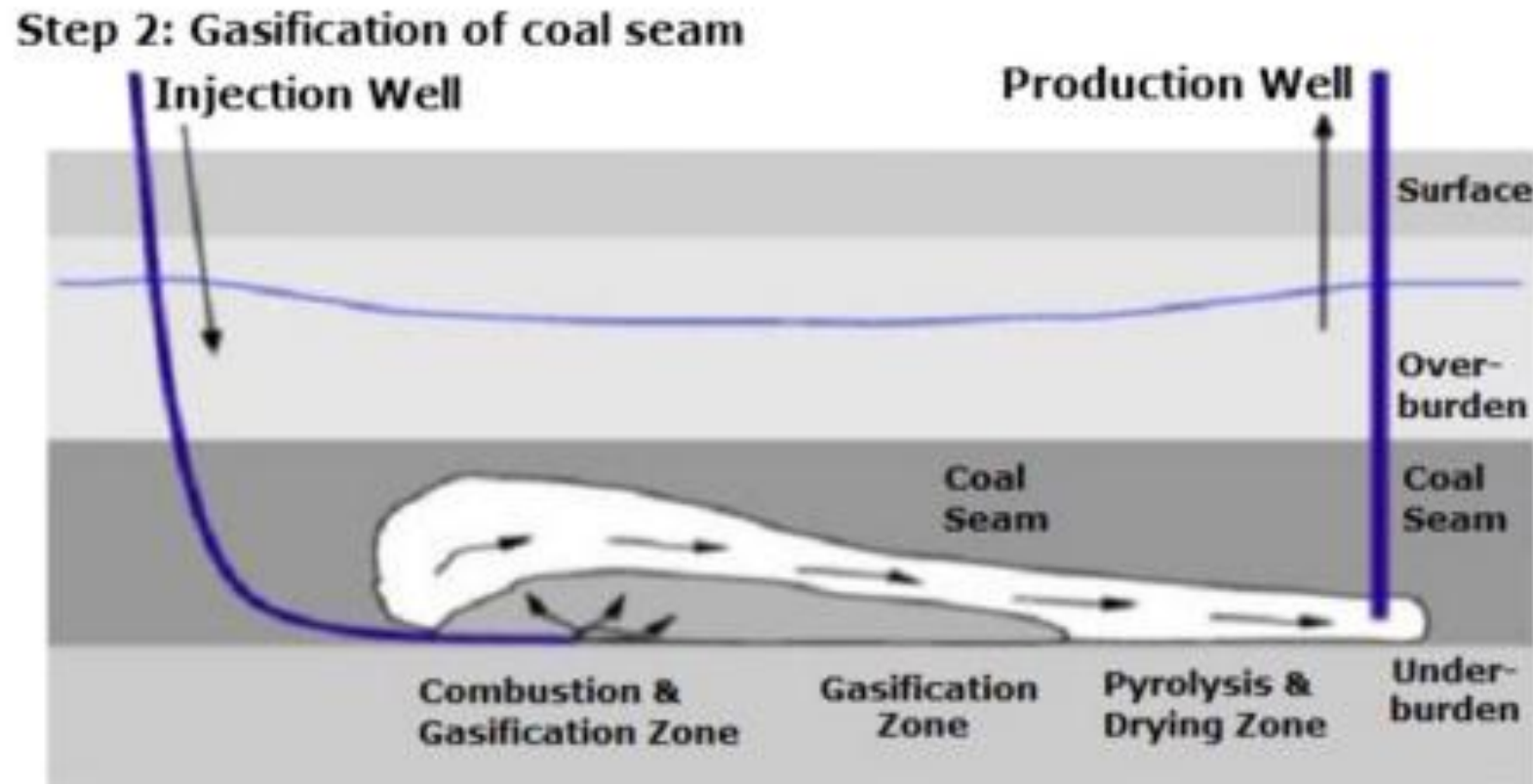
- The product gas requires **cleaning** once it has reached the **surface**, either **to meet the specification** for **input into a gas turbine** (for electricity generation), or to be of sufficient purity for use as a **chemical feedstock** for conversion to **synthetic fuels**.
- The **technique to gasify the coal** in situ uses two vertically drilled wells as the **Injection** and **Production** wells. The procedure consists of three steps.

Step 1: Drill wells and establish permeable link



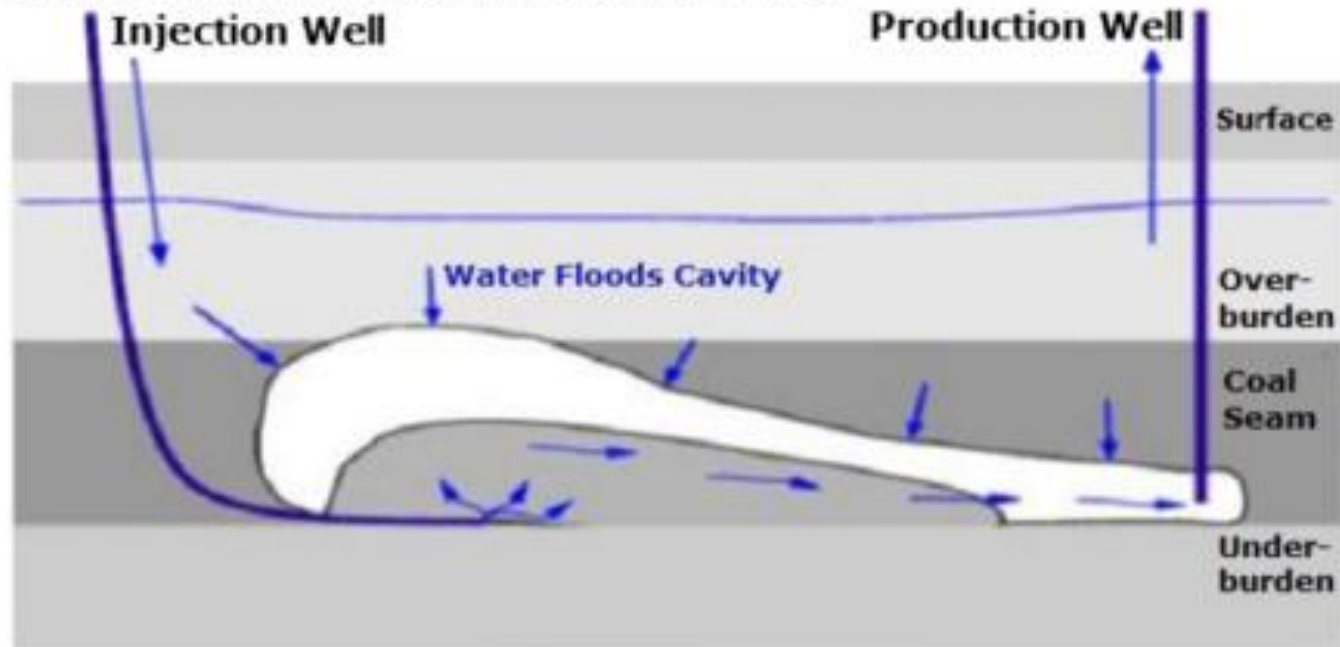
- In the **first step** an **injection and production well** are drilled from the **surface** to the **coal seam** and highly **permeable path** within the coal seams are **established** between these two well.

- Prior to the gasification step a linkage path is created between injector and producer. Several techniques can be used for linking the wells, including the Reverse Combustion Linking (RCL), Forward Combustion Linking (FCL), hydro-fracking, electro-linking, explosive and in-seam linking.



Step 2: coal seam ignition and commencement of gasification

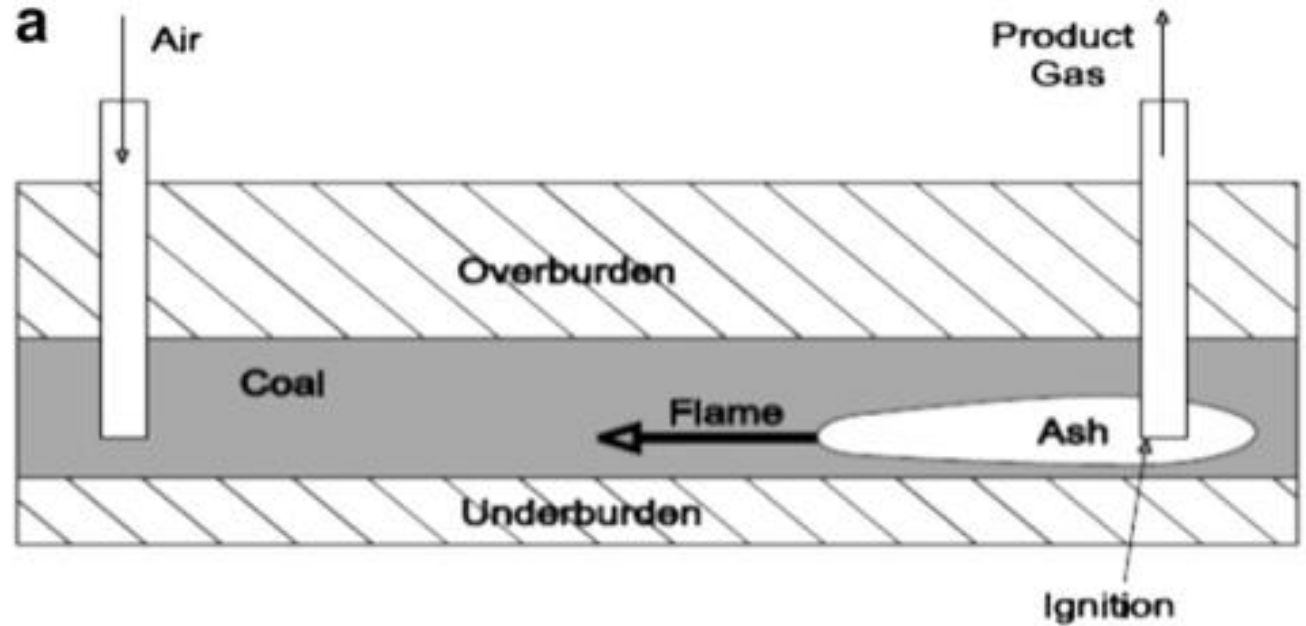
Step 3: Cavity is flushed with steam & water



step 3: site clean-up by flushing cavity with steam and water to remove potential contaminants

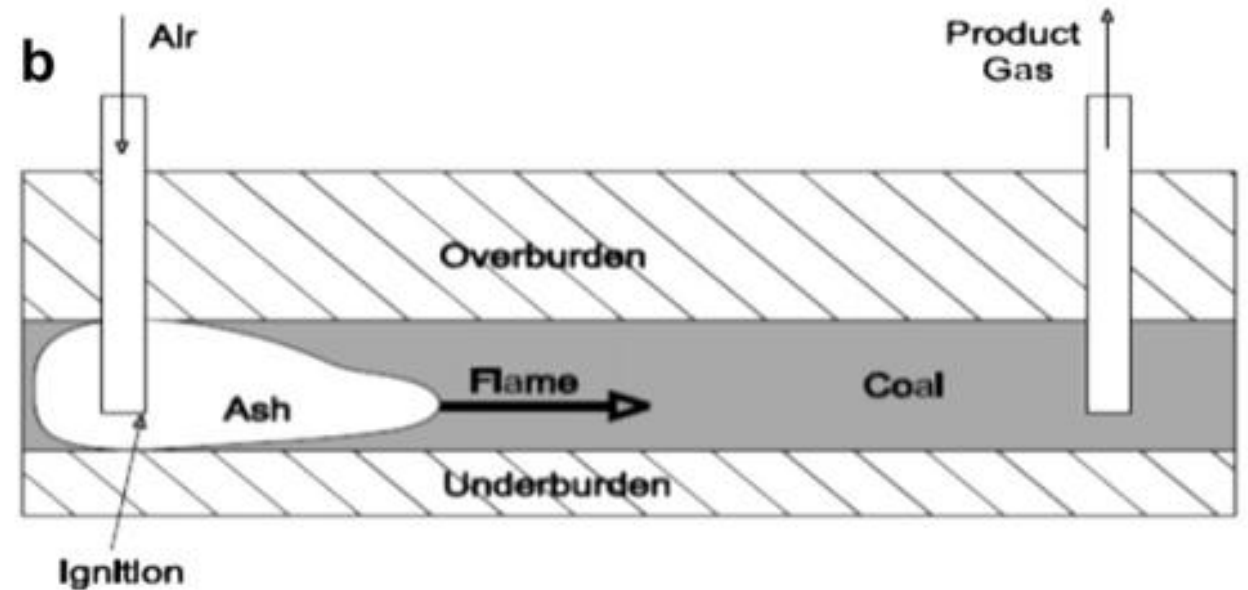
Reverse Combustion Linking (RCL)

- The RCL is a method of linking which includes the injection of an oxidant into one well and the ignition of coal in the other so that combustion propagates toward the source of the oxidant.
- Reverse combustion linking was found to be a reliable method, in relatively shallow and relatively low rank coals (lignite, sub-bituminous coal) and could be operated with well centres between 20 m and 50 m apart.



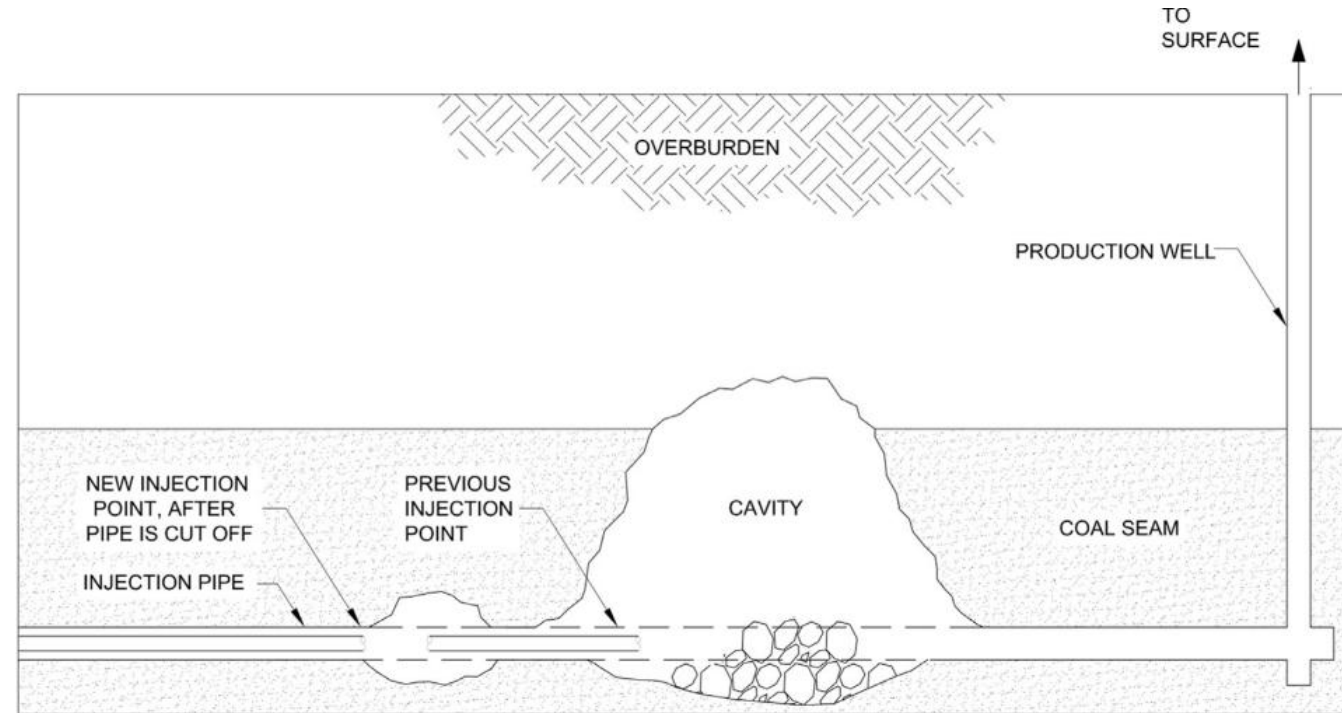
Forward Combustion Linking (FCL)

- In the course of the FCL coal is ignited in the injection well, and the fire propagates toward the production well.
- During forward gasification, the flame-working face gradually moves to the outlet, making the dry distillation zone shorter and shorter.
- At the time when forward gasification is nearly complete, the reduction zone also becomes shorter.



Controlled Retracting Injecting Point (CRIP) Method

- The CRIP technique uses **directional drilling** to **locate an injection well, low and horizontal** in the seam.
- In the first stage, **the injection point is located close to the production well.**
- In the second and subsequent stages the **injection point is retracted away from the production well to form a new gasification cavity in fresh coal.**
- An advantage of the CRIP technique is that injection location remained low in the seam, **injection well damage from subsidence was minimized and large amounts of coal could be gasified with a single injection/production well pair.**



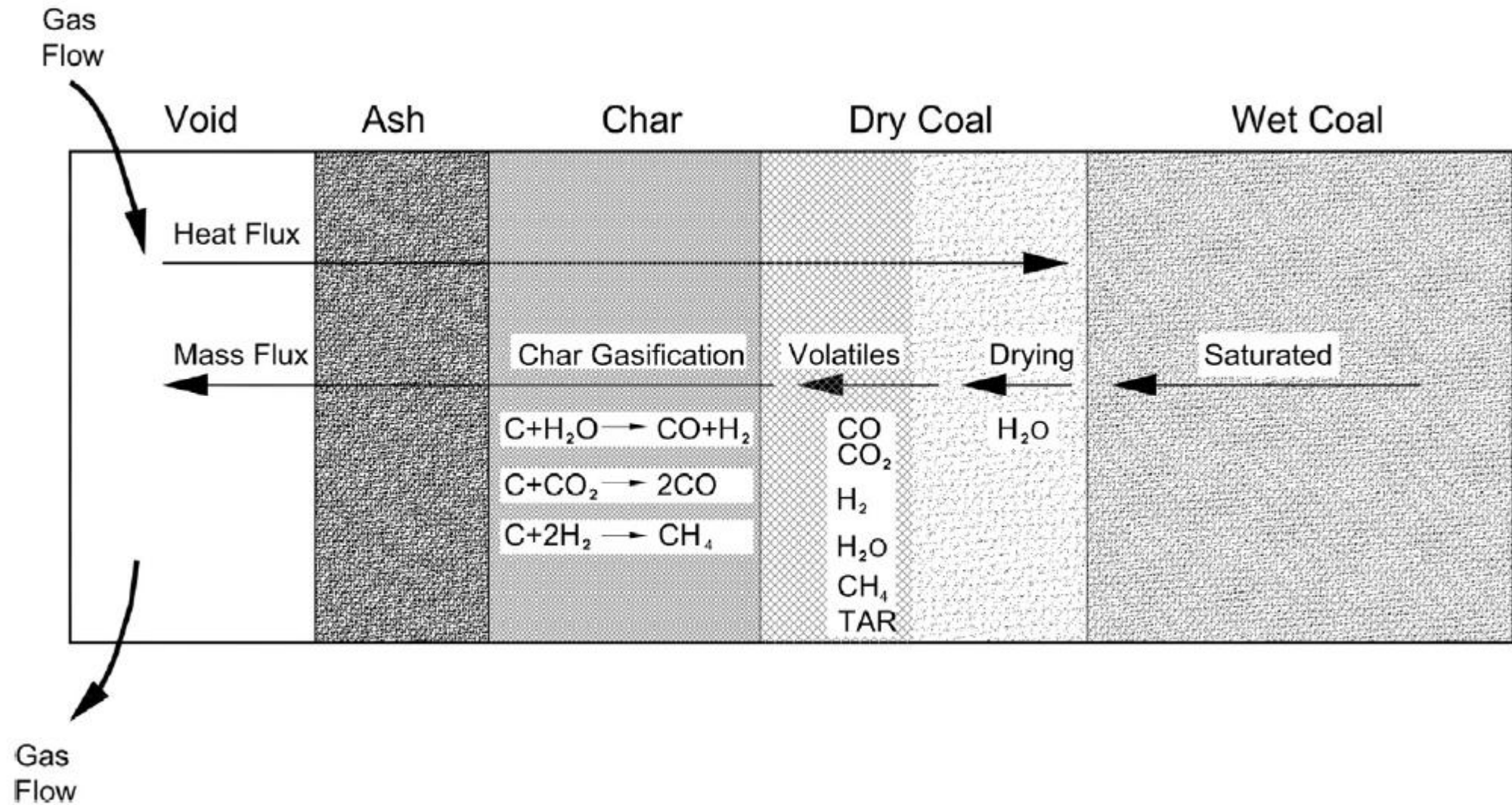
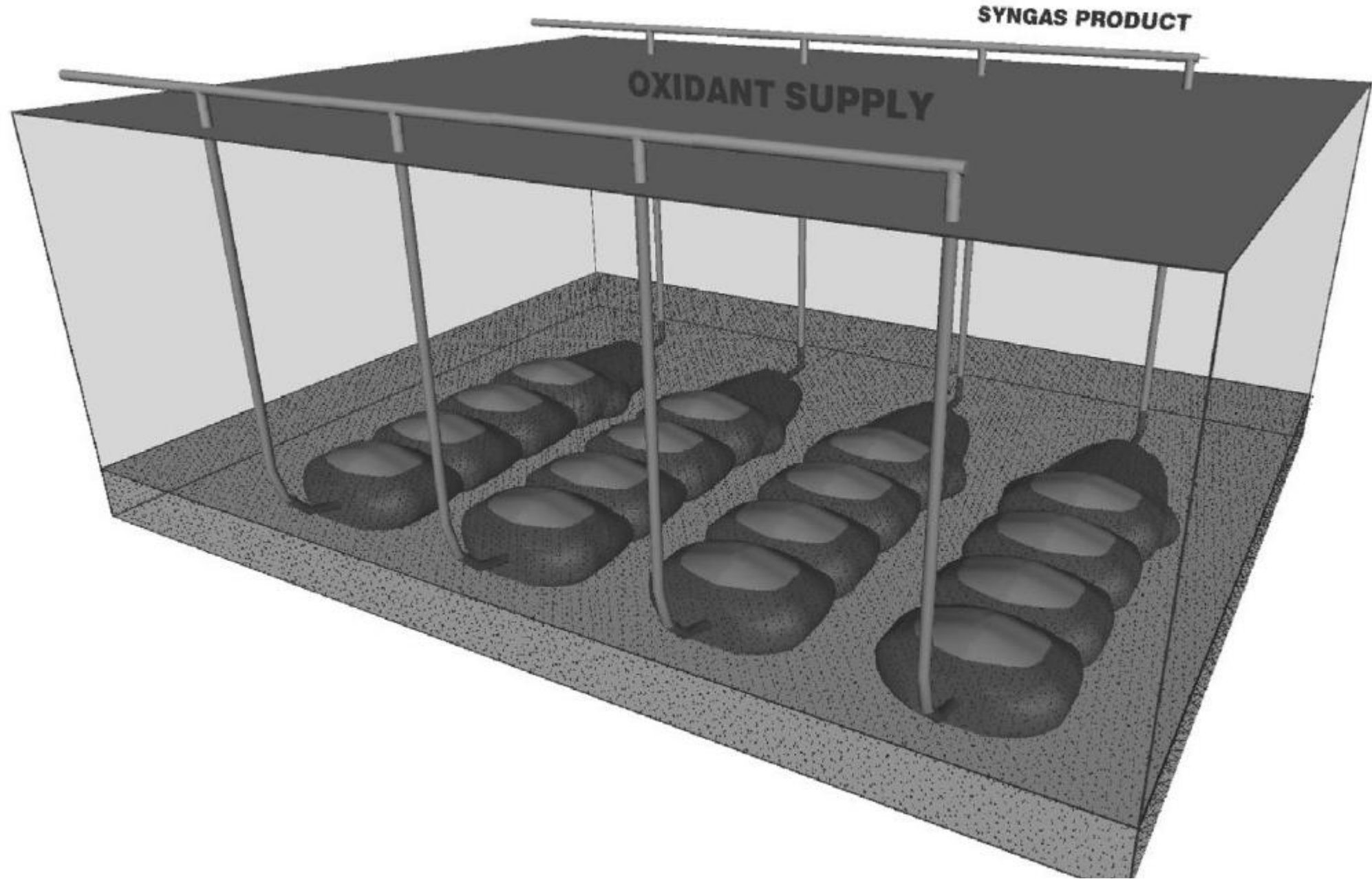
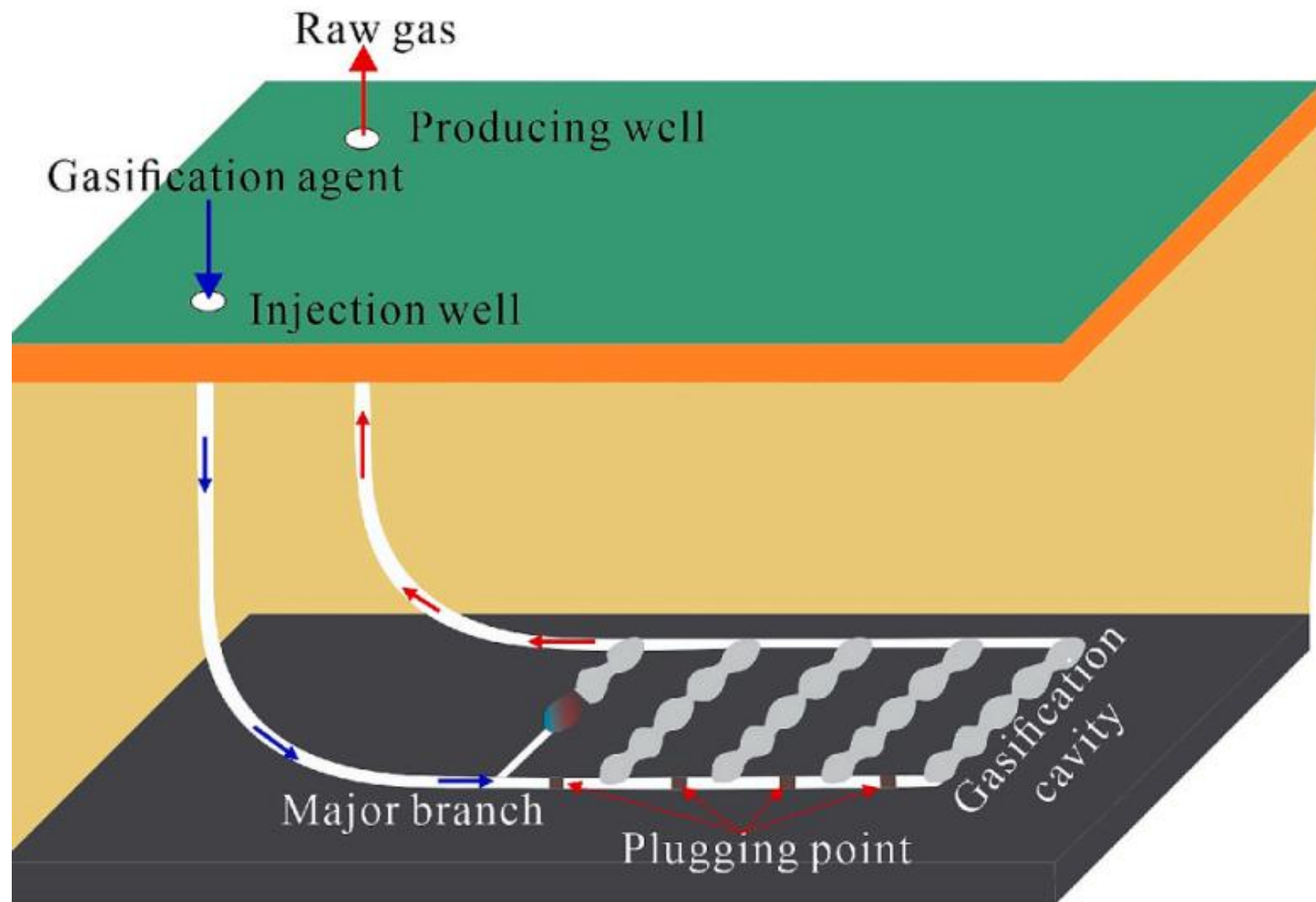


Fig. 24. General representation of the sidewall coal zone of a UCG cavity, showing gas flow in the void, heat transfer into the coal through the ash, char and dry coal layers and the mass flux of syngas and volatile molecules from within the coal into the void space.



General representation of a module of four gasifiers operating simultaneously.

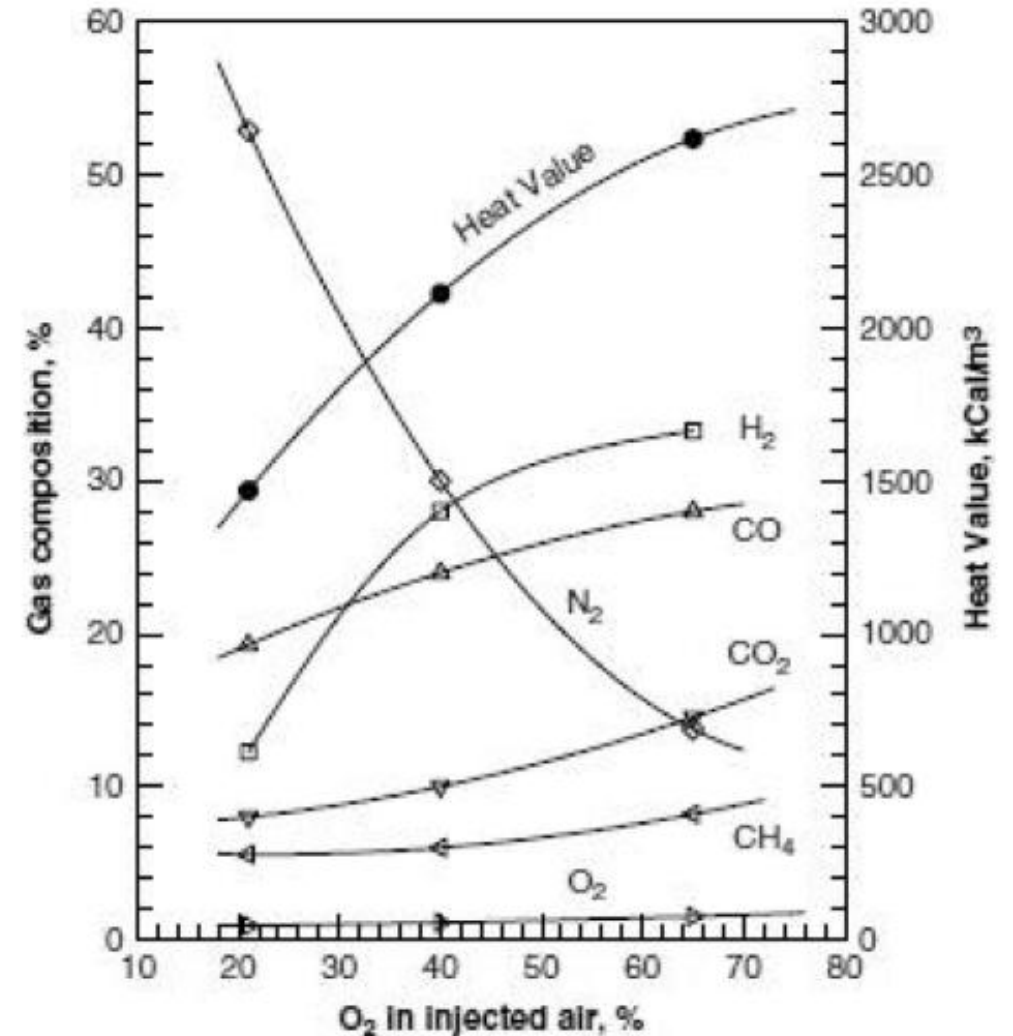


Factors affecting process performance

- Attempts have been made to gasify a wide range of coal types in different coal seam conditions using the discussed techniques.
- Fundamentally, it is desirable to have **low cost, high efficiency, high resource recovery** and reliable operations with **low environmental impacts!**
- In practice there are many decisions and trade-offs to be made in order to achieve a viable process.
- **Effect of oxidant**
- **Effect of coal properties**
- **Effect of coal seam depth**
- **Effect of coal seam thickness**
- **Effect of site conditions**

Effect of oxidant

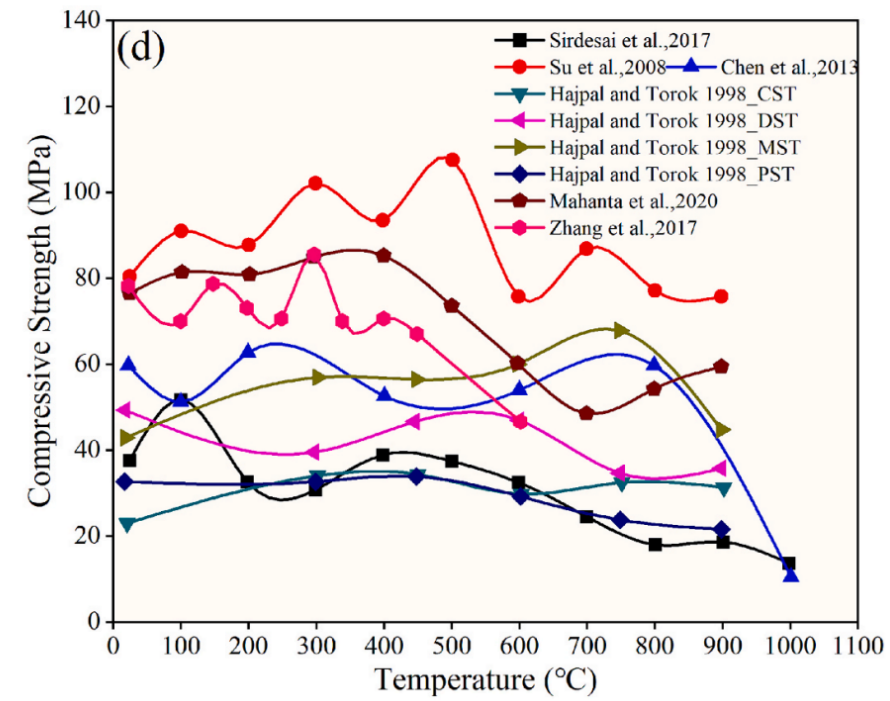
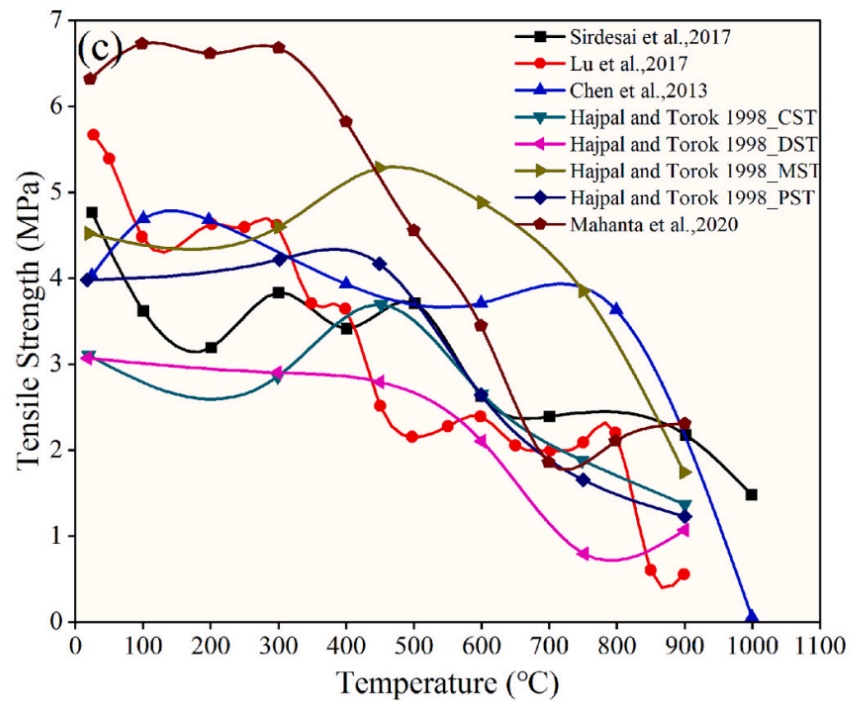
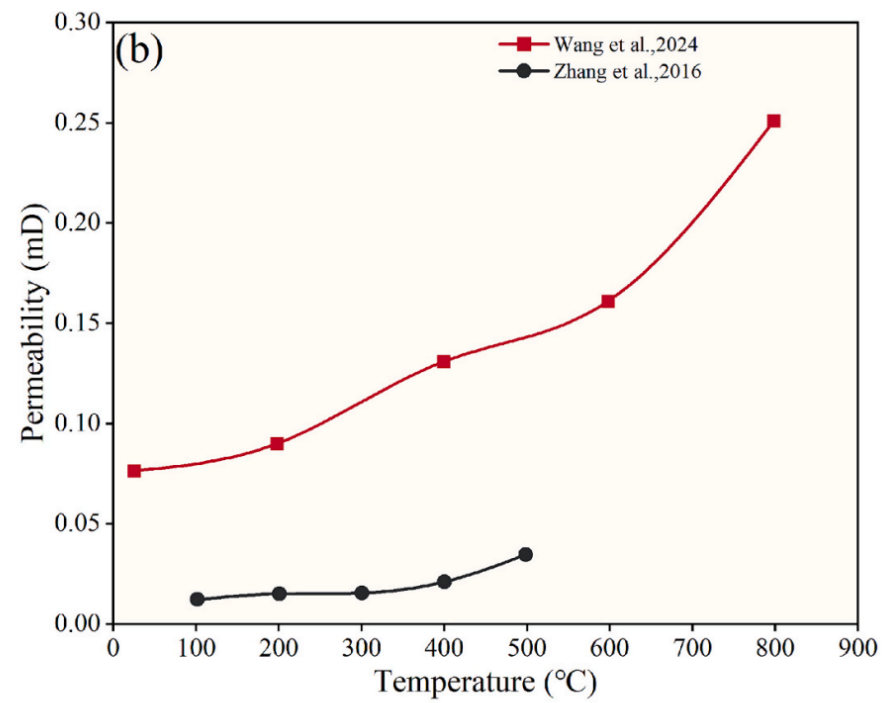
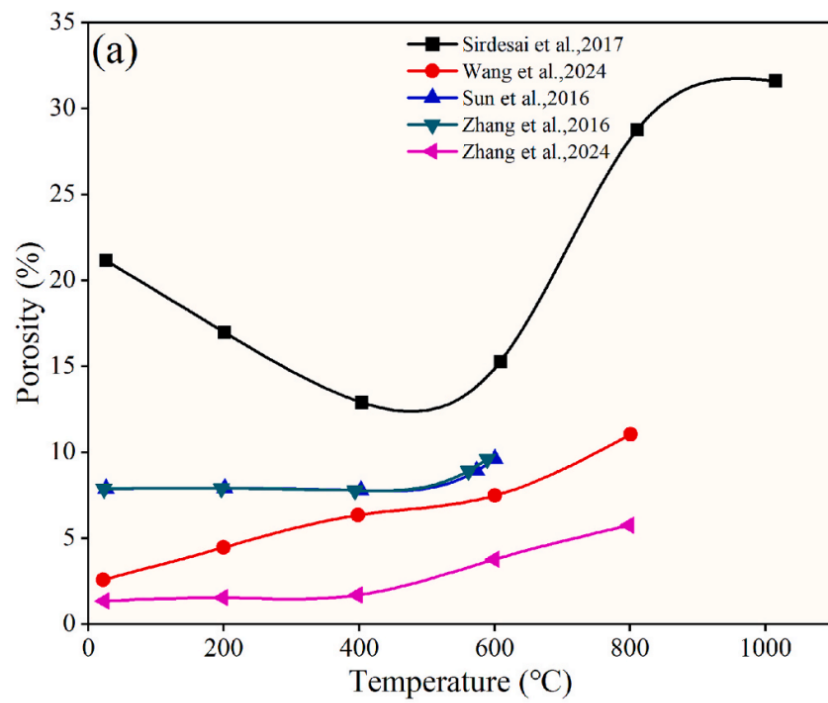
- The injected oxidant is usually either air, enriched air, or a mixture of steam/oxygen or water/oxygen.
- The **rate of gas production** and its **composition** varies widely depending upon the **oxidant used**, **operating conditions** such as **gas injection rate** and **site conditions**, including **type of coal**, and the behaviour of the surrounding **strata** and **hydro-geological** conditions.
- When **air is used**, the product gas has a **lower heating value of 3-6 MJ/m³**. When a **steam/oxygen mixture** is used as the oxidant, the product gas **heating** value can reach **11 MJ/m³** or more.



Effect of Coal Properties

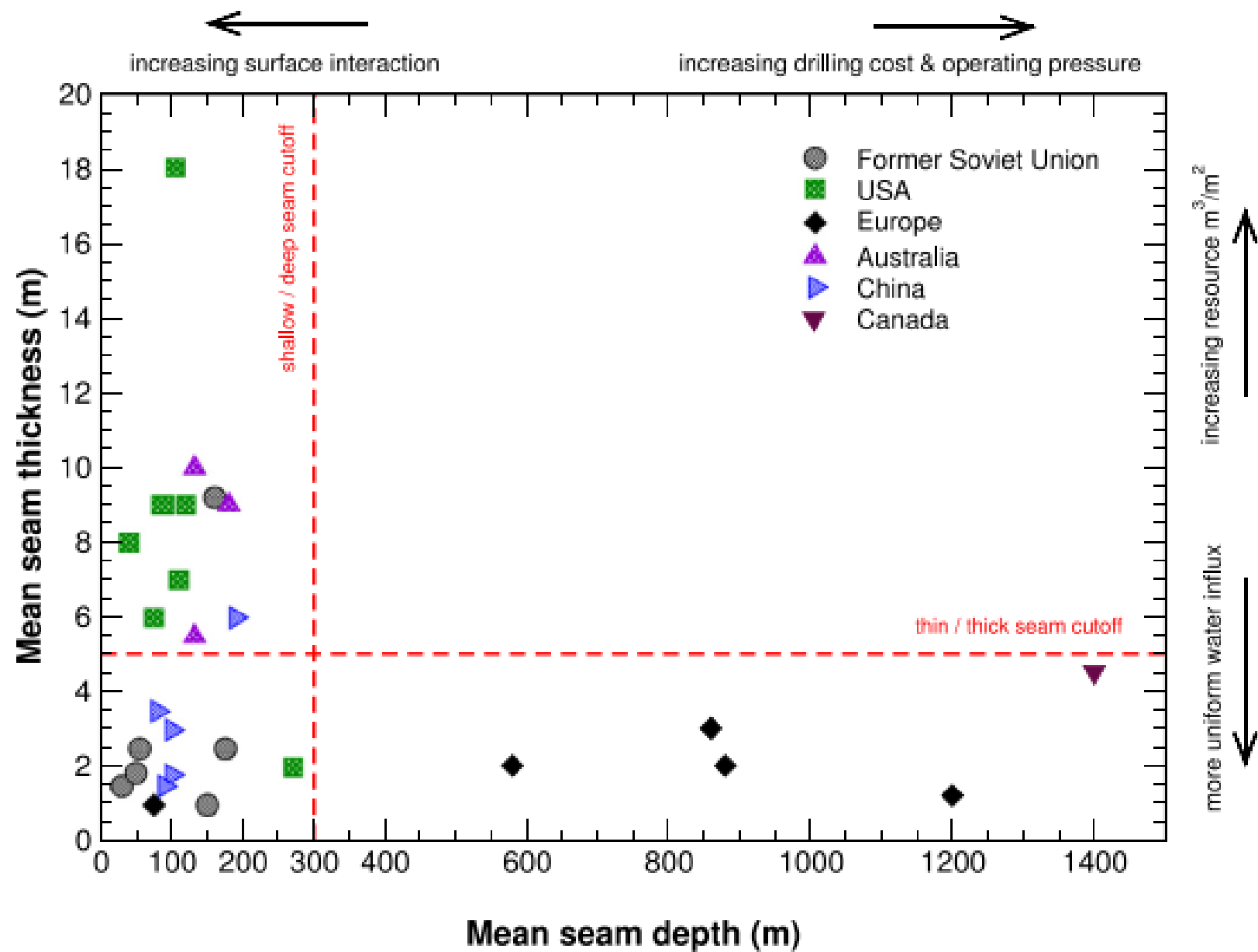
- Coal properties have a substantial effect on the operation and performance of UCG.
- Historically, demonstrations have mostly been conducted in **low-rank coals** which are **reactive** and which **spall upon heating**, which help the **conversion of coal into gas**.
- **The influence of coal rank on the likely performance of UCG**: sub-bituminous coals **give high methane content, moderate to high heating value and a high production rate of gas per unit of coal**. Lignites produce a syngas with lower heating value and produce less gas per ton of coal.
- The rank of the coal, its propensity to spall when heated and the behaviour of mineral matter upon heating are key factors in determining the performance of large scale UCG.

- Low-rank coals generally shrink and spall upon heating, enabling renewal of the sidewall and roof surfaces and the establishment of a permeable bed of coal and ash through which the injected oxidant can readily flow and penetrate to the surface in order to gasify the remaining fixed carbon.
- High rank coals are less reactive and have a tendency to swell upon heating, potentially causing problems with diffusion of volatiles from within the pore structure to reach the surface of the coal.
- In addition, the spalling behaviour of high rank coals is not well understood.
- High rank coals, generally have lower volatile contents than low rank coals, and together with the low reactivity are therefore harder to ignite.



Effect of coal seam depth

- The operating pressure is determined by coal seam depth. As a rule of thumb, the **operating pressure increases by ~ 1 MPa (~ 10 bar) for every 100 m of depth.**
- It can be expected that at shallow depths there is likely to be **increased interaction** between the **cavity** and the **surface** and **operating pressures** will be low.
- As a rule of thumb, UCG should be undertaken in coals deeper than 200 m to ensure sufficient containment pressure.
- As the depth increases, operating pressure increases and so do drilling costs.
- As the pressure increases, there is a general trend towards more CO₂ and CH₄ in the syngas and lower amounts of CO and H₂.



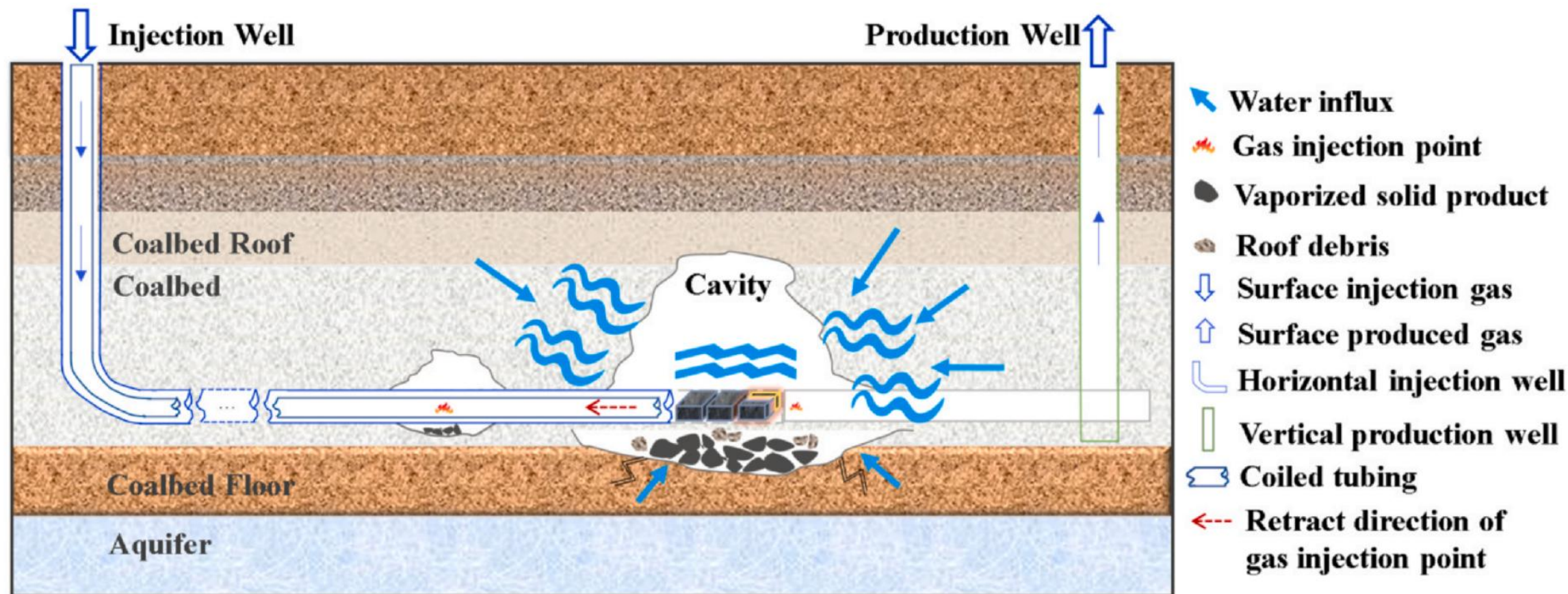


Fig. 2. Physical model of water influx during controlled retracting injection point process.

Effect of coal seam thickness

- Thicker seams offer a proportionately higher potential resource per unit of projected surface area, which should translate into lower costs per unit of coal gasified.
- The syngas heating value is very low ($< 4 \text{ MJ/Nm}^3$) when the coal seam is less than 3 m thick and when the specific water inflow (ton-water/ton-coal-gasified) is greater than 2 ton/ton.
- Thicker coal seams lead to less heat loss per unit volume of coal affected and hence higher syngas heating values.

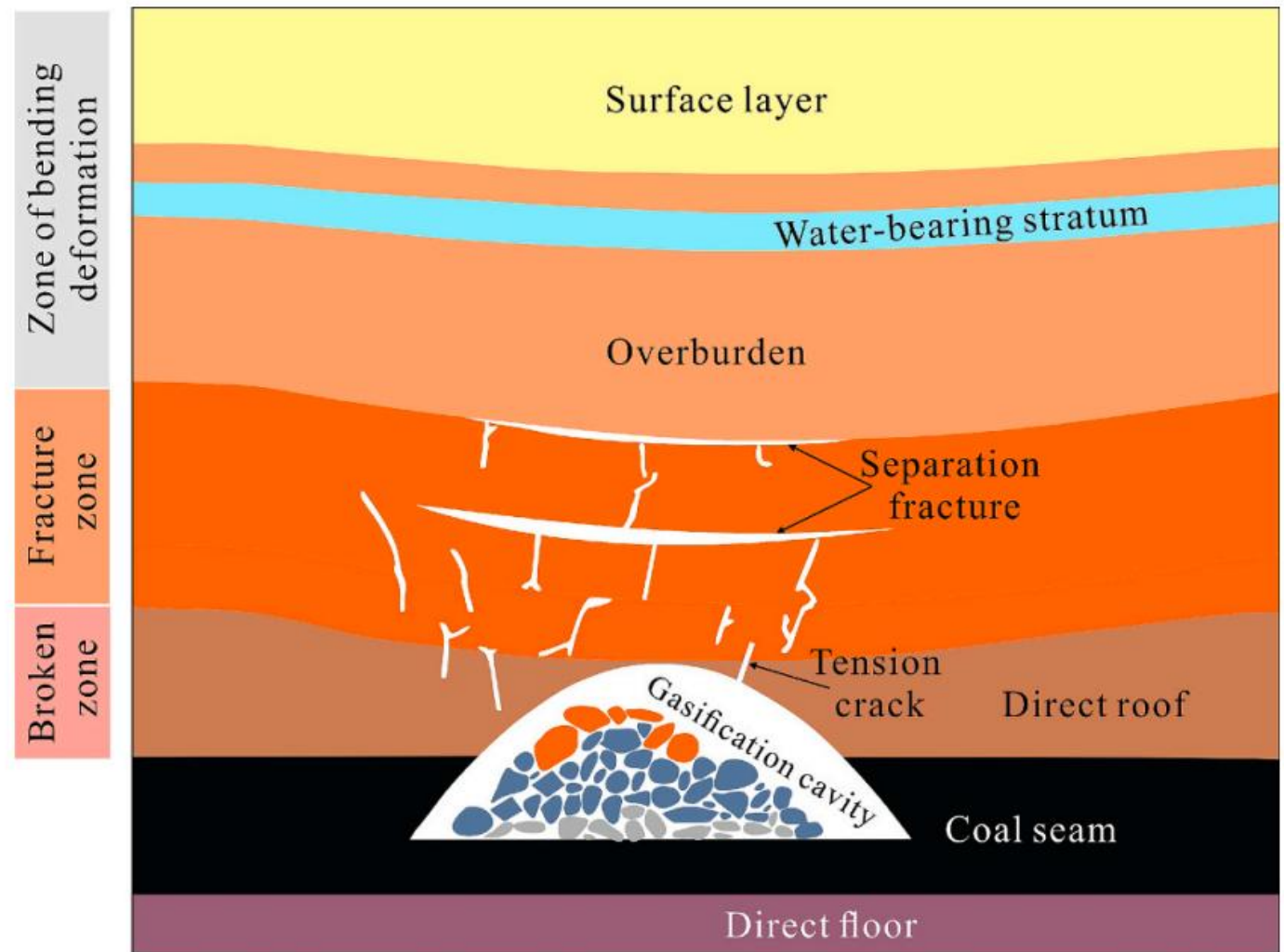
Effect of site conditions

- There are many site factors which influence the process, including **under- and over-burden properties** and **hydro-geological properties** of the formation, in addition to those already considered such as **oxidant and coal properties**.
- Based on the historical data, the more **successful projects** have typically been performed in **relatively shallow coals (< 300 m)**, which are **moderately thick (6-10 m)**, have **low rank** (sub-bituminous coals or lignite), surrounded by **shales, clays or sandstones** and have utilized a **CRIP gasifier** design.
- The **less successful projects** have been performed in either **very shallow or relatively deep** coals, which are **reasonably thin (< 7 m)** and which are of **higher rank**, such as **bituminous coals and anthracites**.

Surface impacts

- UCG operations can lead to the escape of product gases into the air and surface soils and the deformation and subsidence of the surface topography.
- The most likely pathway for product gases to reach the surface air or shallow soils is via the wells (injection, production, monitoring) drilled from the surface.
- Therefore, the design of the casing and cement of the wells is very important.
- Subsidence of the surface ground due to UCG is another potential impact, and it depends heavily on the formation depth (D) and thickness (H), strength of formation materials and the width (W) of the zones of coal conversion.
- The subsidence, δ , from UCM is correlated as $\delta \propto W \cdot H \cdot D^{-1}$

- Surface subsidence has been observed in many UCG projects operated at shallow depth and in moderate to thick coal seams.
- When UCG has been conducted in deeper coal seams with strong over-burden materials there has been negligible surface subsidence.



Advantages of UGC	Limitations
• It can be used to recover the energy content of low rank coals that are not economically or technically feasible to recover by conventional technologies because of their seam thickness, depth, high ash and/or excessive moisture content, large dip angle, or undesirable overburden properties. • Coals that are unmineable (too deep, low grade, thin seams) are exploitable by UGC, thereby greatly increasing resource availability. • Lignite and sub bituminous coal are ideal for gasification, as is bituminous coal provided no significant swelling characteristics exist. • CV's in the range of 12–14 MJ/m³ are recorded when using oxygen feed and this may be slightly increased as the process develops at a greater depth. • Deeper coals offer the opportunity to have much higher pressures in the reactor resulting in higher methane content and resultant higher heat value gas. • The UGC process has a higher thermal efficiency than surface gasification processes. • The CRIP concept has led to the highest gasification efficiency in terms of oxygen usage and will allow subsidence to be minimized or possible eliminated, by using wider barrier pillars between panels. • UGC also lowers the capital investment by eliminating the need for specialized coal mining, processing (transporting and stocking) and gasification reactors, reducing operating costs, surface damage and eliminating mine safety issues such as mine collapse and asphyxiation. • UGC uses less water than surface gasification processes which must maintain a high steam-to-air ratio to avoid slagging. Furthermore, it does not use high quality surface water, but utilizes the water within the coal seam itself. • Most of the ash in the coal stays underground, thereby avoiding the need for excessive gas clean-up, and the environmental issues associated with fly ash waste stored at the surface. • Reclamation of the land is not a serious problem since the surface disturbance is minimal. • There is no production of some criteria pollutants (e.g., SO_x, NO_x) and many other pollutants (mercury, particulates, sulfur species) are greatly reduced in volume and easier to handle. • UGC eliminates much of the energy waste associated with moving waste as well as usable product from the ground to the surface; • UGC, compared to conventional mining combined with surface combustion, produces less greenhouse gas and has advantages for geologic carbon storage. • The well infrastructure for UGC can be used subsequently for geologic CO₂ sequestration operations. It may be possible to store CO₂ in the reactor zone underground as well in adjacent strata. • UGC research and development have been conducted in several countries, including long-term commercial operation of several UGC plants in the former Soviet Union.	• Shallow seams are not suitable for gasification because of high gas losses, potential breakthrough to surface and possible contamination of groundwater. The shallowness of the coal seams means the pressure in the reactor is low and the quality of the gas produced is reduced compared to deeper systems. Heat losses are considerable with such seams, leading to low thermal efficiency and lower product gas quality. • Thin seams, less than 3 m thick, will be difficult to exploit economically unless in a multi-seam environment. • Installation of well pairs (injection and production wells) is costly and therefore it is desirable to gasify the maximum volume of coal between a well pair. • During UGC processes, the surrounding rock acting as the furnace walls will be affected by high temperature, and its mechanical properties will change with the increased temperatures. • Stress and displacement will happen among rocks due to the high temperature.